



Data Publishing Policy

Version 2 - Updated June 2025

Background

The Pathogen Data Network (PDN) recognizes the importance of rapidly and widely sharing data during disease outbreaks. The rapid, open, and wide sharing of data on emerging pathogens helps keep people and the planet healthy.

PDN believes in the power of open data to a) facilitate public health responses to pathogenic emergencies and b) to advance the scientific research agenda which, in turn, furthers future pandemic preparedness and prevention. For these reasons, PDN is putting forth data sharing guidelines.

For whom is this policy intended?

These guidelines are meant to inform the work of the PDN consortium members, those operating Pathogen Portal Nodes, and others who wish to share their data through the Pathogens Portal.

What is the policy?

During outbreaks, the kinds of data that need to be shared and the speed by which they should be shared are different from what should occur during non-outbreak times for surveillance and academic research. For this reason, we have different data sharing protocols/requests for different scenarios. We also recognize that onerous metadata standards might discourage uploading-- we hope to help submitters upload as much good quality, necessary metadata as is feasible; but if there is a choice between publishing data with sub-par metadata or not publishing at all, we advocate for publishing. We also have policies for ensuring proper credit and attribution for data generators/uploaders.

In general, PDN data -- that is data which can be found through the Pathogens Portal -- is open and freely accessible. As more deposits are made into databases and repositories from which the Portal draws, more data will become available openly. PDN encourages data repositories to make their submission processes as straightforward as possible.

During Outbreaks, to facilitate a rapid public health response, PDN expects our partners to rapidly (within days or weeks) share:

- Any permissible and privacy protecting epidemiological data (time series, geographical location, demographic information) on the ongoing outbreak
- Full sequence data, if possible, including methods and protocols

- Annotations from the analyses of genomic data (e.g., presence of specific mutations, lineage, etc.)

We ask that these data are published openly, without prejudice to national or international laws.

During the course of normal disease surveillance, we encourage our partners to share aggregate, privacy protecting surveillance data (epidemiological, sequence, and annotation data) in regular intervals (every 3-6 months), if it is possible to do so given national contexts. In particular, we request the publishing of data on priority pathogens.

During the course of normal pathogen-related research, we ask that our partners make all their pathogen-related data at minimum discoverable by making their metadata catalogues (similar data types to epidemiological data and information about data owner, availability of data, etc.) available as soon as possible. The more data is available from the course of normal research, the more prepared the world is to respond to the next outbreak.

We encourage our partners to eventually (6 months- 1 year) make all their data (raw sequence data, fully assembled sequences, annotations, permissible metadata) openly available through deposition into an open access global repository or database.

How will attribution work?

Attribution to data generators/uploaders is a critical part of data sharing, and those submitting to the Pathogens Portal will find fields in which data attribution (e.g., ORCIDs) can be added. We support academics and researchers getting credit for inputting to the global data commons, and think that linking data to ORCIDs and requesting that data users cite their data in publications can go a long way to giving credit to researchers. We also strongly encourage data users' collaborations with identified data generators, when feasible. We will also look into text fields where data uploaders can note their funders.

Areas for further work

At present, this document is meant to serve as a general guideline for data sharing within the PDN. As PDN evolves, this document should either link to more information or contain more information on the following topics:

- Specific minimum viable metadata standards for different scenarios (epidemiological and technological)
- Information about what specific sequence data is necessary and when
- Information about what 'privacy protecting' data sharing looks like in different national and epidemiological contexts
- Information on dealing with restrictive national legal requirements.